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On strong ergodicity for nonhomogeneous continuous-time Markov chains

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Let $X(t)$ be a nonhomogeneous continuous-time Markov chain. Suppose that the intensity matrices of $X(t)$ and some weakly or strongly ergodic Markov chain $\bar{X}(t)$ are close. Some sufficient conditions for weak and strong ergodicity of $X(t)$ are given and estimates of the rate of convergence are proved. Queue-length for a birth and death process in the case of asymptotically proportional intensities is considered as an example.

nonhomogeneous * continuous-time Markov chain * strong ergodicity * weak ergodicity * forward Kolmogorov system * differential equation in the space l_1

1. Introduction and definitions

Ergodic properties of nonhomogeneous continuous-time Markov chains have been investigated by a number of authors. One of the problems in this area is the following. Let the intensity matrix of a Markov chain $X(t)$ be close to the intensity matrix of an ergodic Markov chain $\bar{X}(t)$ (particularly, the perturbation tends to zero as $t \to \infty$ or the perturbation is integrable). Is $X(t)$ also ergodic with the same limit regime?

The case where $\bar{X}(t)$ is homogeneous was considered in [6], and the case of an exponential ergodic 'non-perturbed' chain was studied in [12]. Now we consider the more general case and give some estimates of the rate of convergence.

Let $X(t)$ and $\bar{X}(t)$ be continuous-time Markov chains with the state space $S = \{0, 1, \ldots, K\}$, $K \leqslant \infty$. Write $p_i(t) = \Pr(X(t) = i)$ and $\boldsymbol{p}(t) = (p_i(t))$. Let $P(s, t) = (p_{ij}(s, t))$ and $A(t) = (a_{ij}(t))$ be transition and intensity matrices for the chain $X(t)$ respectively.

Put $\|x\| = \sum |x_i|$, $\|P\| = \sup_i \sum_j |p_{ij}|$; let Ω be the set of all stochastic vectors, and $'$ denote transpose.

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We shall consider the Markov chains which satisfy the following condition: for any $t \geqslant 0$ and $h > 0$,

$$\Pr\{X(t+h) = j / X(t) = i\} = \begin{cases} a_{ij}(t)h + o(h) & \text{if } j \neq i, \\ 1 - h \sum\limits_{k \neq i} a_{ik}(t) + o(h) & \text{if } j = i, \end{cases} \tag{1.1}$$

where all $o(h)$ are uniform in i, $a_{ii}(t) = -\sum_{k \neq i} a_{ik}(t)$. Then for any $t \geqslant 0$ in l_1-norm,

$$\boldsymbol{x}(t+h) - \boldsymbol{x}(t) = B(t)\boldsymbol{x}(t)h + o(h), \tag{1.2}$$

where $\boldsymbol{x}(t) = \boldsymbol{p}'(t)$, $B(t) = A'(t)$. Hence there is the following differential equation in the space l_1:

$$d\boldsymbol{x}/dt = B(t)\boldsymbol{x}. \tag{1.3}$$

One can note that this equation is a forward Kolmogorov system for $X(t)$. If $U(t, s)$ is a Cauchy operator of equation (1.3) then for any $s \geqslant 0$, $t \geqslant s$, $U(t, s) = P'(s, t)$. Notice that

$$U(t, \tau) = I + \int_\tau^t B(s)\, ds + \int_\tau^t B(s) \int_\tau^s B(s_1)\, ds_1\, ds + \cdots \tag{1.4}$$

and $\boldsymbol{x}(t) = U(t, s)\boldsymbol{x}(s)$ for any $s \geqslant 0$, $t \geqslant s$, $\boldsymbol{x}(s)$; the properties of a Cauchy operator were investigated in [1].

Denote by $\bar{p}_i(t), \bar{\boldsymbol{p}}(t), \bar{P}(s, t), \bar{A}(t)$ the respective functions for the chain $\bar{X}(t)$.

In the whole paper we suppose that the intensity matrices are locally integrable on $[0, \infty)$ and the forward Kolmogorov system is identified with a differential equation in the space l_1, see [1,10]. In this case Cauchy operators for $X(t)$ and $\bar{X}(t)$ exist and are unique.

Definition 1.1. A Markov chain $X(t)$ is called weakly ergodic if there exists a vector function $\boldsymbol{q}(t) \in \Omega$ for every $t \geqslant 0$ such that if $Q(t) = (\boldsymbol{q}(t), \boldsymbol{q}(t), \ldots)'$ then for all $s \geqslant 0$,

$$\lim_{t \to \infty} \| P(s, t) - Q(t) \| = 0. \tag{1.5}$$

It is noteworthy that $X(t)$ can be proved to be weakly ergodic if the above limit is equal to zero for some $s \geqslant 0$.

Definition 1.2. A Markov chain $X(t)$ is called strongly ergodic if there exists a vector $\boldsymbol{q} \in \Omega$ such that if $Q = (\boldsymbol{q}, \boldsymbol{q}, \ldots)'$ then for all $s \geqslant 0$,

$$\lim_{t \to \infty} \| P(s, t) - Q \| = 0. \tag{1.6}$$

Definition 1.3. A Markov chain $X(t)$ is called exponentially ergodic if there exist positive numbers N and σ such that for any $\boldsymbol{p}^* \in \Omega, \boldsymbol{p}^{**} \in \Omega$, s, t $(0 \leqslant s \leqslant t)$,

$$\| (\boldsymbol{p}^* - \boldsymbol{p}^{**}) P(s, t) \| \leqslant N e^{-\sigma(t-s)} \| \boldsymbol{p}^* - \boldsymbol{p}^{**} \| . \tag{1.7}$$

Let

$$a(t) = \| A(t) - \bar{A}(t) \| . \tag{1.8}$$

Then using [6,12] we have the following:

Proposition. *If $\bar{X}(t)$ is strong ergodic and homogeneous (or exponentially ergodic) and, in addition, $a(t) \to 0$ as $t \to \infty$, then $X(t)$ is strongly ergodic and the limit regimes for $X(t)$ and $\bar{X}(t)$ are identical.* □

Remark 1.1. The proposition is not correct if $\bar{X}(t)$ is assumed to be only strongly ergodic.

Example 1.1. Let $X(t)$, $\bar{X}(t)$ be Markov chains with two states 0, 1; and let $\lambda(t)$, $\mu(t)$, $(\bar{\lambda}(t), \bar{\mu}(t))$ be intensities of transitions $0 \to 1$ and $1 \to 0$ respectively, where $\bar{\lambda}(t) = \bar{\mu}(t) = 1/(1+t)$, $\lambda(t) = \mu(t) \equiv 0$. Then $\bar{X}(t)$ is strongly ergodic with the limit regime $\boldsymbol{q} = (0.5, 0.5)$, $a(t) = 2/(1+t) \to 0$ as $t \to \infty$; and $X(t)$ is not ergodic.

On the other hand the condition of exponential ergodicity of $\bar{X}(t)$ is not necessary; for instance, if $\bar{\lambda}(t) = \bar{\mu}(t) = 1/(1+t)$, $a(t) = 1/(1+t)^2$, then the chain $X(t)$ in Example 1.1 is strongly ergodic with the limit regime $\boldsymbol{q} = (0.5, 0.5)$.

The following definition seems to be applicable for the study of ergodicity of 'perturbed' Markov chains. Let $N > 0$ and let $\alpha(t)$ be a non-negative locally integrable function on $[0, \infty)$ with

$$\int_0^\infty \alpha(t)\, \mathrm{d}t = \infty . \tag{1.9}$$

Definition 1.4. A Markov chain $X(t)$ is $(N, \alpha(t))$-ergodic if for any $\boldsymbol{p}^* \in \Omega, \boldsymbol{p}^{**} \in \Omega$ and all s, t $(0 \leqslant s \leqslant t)$ the following inequality is satisfied

$$\| (\boldsymbol{p}^* - \boldsymbol{p}^{**}) P(s, t) \| \leqslant N \exp\left(- \int_s^t \alpha(u)\, \mathrm{d}u \right) \| \boldsymbol{p}^* - \boldsymbol{p}^{**} \| . \tag{1.10}$$

If a Markov chain is homogeneous and $(N, \alpha(t))$-ergodic then it is exponentially strongly ergodic. In the general case $(N, \alpha(t))$-ergodicity of a Markov chain implies its weak ergodicity. One can note that the chain $\bar{X}(t)$ in Example 1.1 is $(N, \alpha(t))$-ergodic with $N = 1$, $\alpha(t) = \bar{\lambda}(t) + \bar{\mu}(t) = 2/(1+t)$.

2. A stability theorem

We consider the following condition for a 'perturbation' of a Markov chain $\bar{X}(t)$ when $\bar{X}(t)$ is $(N, \alpha(t))$-ergodic:

$$\text{for almost all } t \geqslant 0, \quad a(t) \leqslant \omega(t)\alpha(t)\,, \tag{2.1}$$

where $\omega(t) \to 0$ as $t \to \infty$.

Theorem 2.1. *Let a Markov chain $\bar{X}(t)$ be $(N, \alpha(t))$-ergodic and the perturbed chain $X(t)$ be such that condition (2.1) is fulfilled. Then for all s, t $(0 \leqslant s \leqslant t)$ the following inequality holds:*

$$\|P(s, t) - \bar{P}(s, t)\| \leqslant 12N \int_s^t \omega(\tau)\alpha(\tau) \exp\left(-\int_\tau^t \alpha(\xi)\,\mathrm{d}\xi\right)\mathrm{d}\tau\,. \tag{2.2}$$

Proof. Let $\boldsymbol{x}(t) = \boldsymbol{p}'(t)$, $\bar{\boldsymbol{x}}(t) = \bar{\boldsymbol{p}}'(t)$, $B(t) = A'(t)$, $\bar{B}(t) = \bar{A}'(t)$. Then $U(t, s) = P'(s, t)$, $\bar{U}(t, s) = \bar{P}'(s, t)$ are the Cauchy operators of the forward Kolmogorov systems:

$$\mathrm{d}\boldsymbol{x}/\mathrm{d}t = B(t)\boldsymbol{x}\,, \qquad \mathrm{d}\bar{\boldsymbol{x}}/\mathrm{d}t = \bar{B}(t)\bar{\boldsymbol{x}}\,, \tag{2.3}$$

respectively.

Put $p_0 = 1 - \sum_{i \geqslant 1} p_i$; then from the forward Kolmogorov systems for the chains $X(t)$ and $\bar{X}(t)$ we have the equations

$$\mathrm{d}z/\mathrm{d}t = B_1(t)z + \boldsymbol{f}(t) \tag{2.4}$$

and

$$\mathrm{d}\bar{z}/\mathrm{d}t = \bar{B}_1(t)\bar{z} + \bar{\boldsymbol{f}}(t)\,, \tag{2.5}$$

respectively. Here $z = (p_1, p_2, \ldots)'$, $\boldsymbol{f} = (a_{01}, a_{02}, \ldots)'$, $B_1 = (b_{1ij})$, $i, j \geqslant 1$, $b_{1ij} = a_{1ji} = a_{ji} - a_{0i}$, and $\bar{z}, \bar{\boldsymbol{f}}, \bar{B}_1$ are the corresponding functions for the chain $\bar{X}(t)$.

The equation (2.4) can be rewritten in the following form:

$$\begin{aligned}\mathrm{d}z/\mathrm{d}t &= \bar{B}_1(t)z + \bar{\boldsymbol{f}}(t) + (B_1(t) - \bar{B}_1(t))z + (\boldsymbol{f}(t) - \bar{\boldsymbol{f}}(t)) \\ &= \bar{B}_1(t)z + \bar{\boldsymbol{f}}(t) + \hat{B}_1(t)z + \hat{\boldsymbol{f}}(t)\,,\end{aligned} \tag{2.6}$$

where $\hat{B}_1(t) = B_1(t) - \bar{B}_1(t)$, $\hat{\boldsymbol{f}}(t) = \boldsymbol{f}(t) - \bar{\boldsymbol{f}}(t)$.

Let $V(t, s)$ be a Cauchy operator for the differential equation

$$\mathrm{d}\boldsymbol{y}/\mathrm{d}t = \bar{B}_1(t)\boldsymbol{y}\,. \tag{2.7}$$

Then known formulae (see, for example, [1]) with $\bar{z}(s) = z(s)$ imply

$$\bar{z}(t) = V(t, s)\bar{z}(s) + \int_s^t V(t, \tau)\bar{\boldsymbol{f}}(\tau)\,\mathrm{d}\tau\,, \tag{2.8}$$

$$z(t) = V(t, s)\bar{z}(s) + \int_s^t V(t, \tau)\bar{f}(\tau)\,\mathrm{d}\tau + \int_s^t V(t, \tau)(\hat{B}_1(\tau)z(\tau) + \hat{f}(\tau))\,\mathrm{d}\tau$$

$$= \bar{z}(t) + \int_s^t V(t, \tau)(\hat{B}_1(\tau)z(\tau) + \hat{f}(\tau))\,\mathrm{d}\tau. \tag{2.9}$$

Therefore

$$\|z(t) - \bar{z}(t)\| \leqslant \int_s^t \|V(t, \tau)\|(\|\hat{B}_1(\tau)\|\|z(\tau)\| + \|\hat{f}(\tau)\|)\,\mathrm{d}\tau. \tag{2.10}$$

We have

$$\|\hat{B}_1(t)\| \leqslant 2a(t) \leqslant 2\omega(t)\alpha(t), \tag{2.11a}$$

$$\|\hat{f}(t)\| \leqslant a(t) \leqslant \omega(t)\alpha(t), \tag{2.11b}$$

$$\|z(t)\| \leqslant \|x(t)\| = \|p(t)\| = 1. \tag{2.11c}$$

Now one can estimate $\|V(t, \tau)\|$. Notice that

$$x = \begin{pmatrix} 1 - \|z\| \\ z \end{pmatrix}. \tag{2.12}$$

It implies the inequality

$$\|z^* - z^{**}\| \leqslant \|p^* - p^{**}\| \leqslant 2\|z^* - z^{**}\|. \tag{2.13}$$

Then one has from (2.13) and (1.10) the following estimate: For any s, t $(0 \leqslant s \leqslant t)$ and any $z^*(s) = z^*$, $z^{**}(s) = z^{**}$,

$$\|V(t, s)(z^* - z^{**})\|$$

$$= \|z^*(t) - z^{**}(t)\| \leqslant \|p^*(t) - p^{**}(t)\|$$

$$= \|(p^* - p^{**})P(s, t)\| \leqslant N\exp\left(-\int_s^t \alpha(\tau)\,\mathrm{d}\tau\right)\|p^* - p^{**}\|$$

$$\leqslant 2N\exp\left(-\int_s^t \alpha(\tau)\,\mathrm{d}\tau\right)\|z^* - z^{**}\|. \tag{2.14}$$

Hence for all t, s $(0 \leqslant s \leqslant t)$,

$$\|V(t, s)\| \leqslant 2N\exp\left(-\int_s^t \alpha(\tau)\,\mathrm{d}\tau\right). \tag{2.15}$$

Now using (2.15) and (2.11) we have from the estimate (2.10) the following inequality:

$$\|z(t)-\bar{z}(t)\| \leqslant 6N \int_s^t \omega(\tau)\alpha(\tau)\exp\left(-\int_\tau^t \alpha(\xi)\,\mathrm{d}\xi\right)\mathrm{d}\tau. \tag{2.16}$$

Inequalities (2.13) and (2.16) imply (2.2). □

Remark 2.1. One can prove that estimate (2.2) has the exact order.

3. Ergodic properties

Theorem 3.1. *Let $\bar{X}(t)$ be $(N, \alpha(t))$-ergodic with limit regime $Q(t)$ and let a perturbed chain $X(t)$ is such that (2.1) is fulfilled. Then $X(t)$ is weakly ergodic and*

$$\|P(s,t)-Q(t)\| \leqslant 2N\exp\left(-\int_s^t \alpha(\tau)\,\mathrm{d}\tau\right) \times\left\{1+6\int_s^t \omega(\tau)\alpha(\tau)\exp\left(\int_s^\tau \alpha(\xi)\,\mathrm{d}\xi\right)\mathrm{d}\tau\right\}. \tag{3.1}$$

Proof. Using inequality (2.15) and Theorem 2.1 one has the following estimate:

$$\begin{aligned}
&\|P(s,t)-Q(t)\| \\
&\quad\leqslant \|P(s,t)-\bar{P}(s,t)\| + \|\bar{P}(s,t)-Q(t)\| \\
&\quad\leqslant 12N\int_s^t \omega(\tau)\alpha(\tau)\exp\left(-\int_\tau^t \alpha(\xi)\,\mathrm{d}\xi\right)\mathrm{d}\tau + 2N\exp\left(-\int_s^t \alpha(\tau)\,\mathrm{d}\tau\right) \\
&\quad\leqslant 2N\exp\left(-\int_s^t \alpha(\tau)\,\mathrm{d}\tau\right) \\
&\qquad\times\left\{1+6\int_s^t \omega(\tau)\alpha(\tau)\exp\left(\int_s^\tau \alpha(\xi)\,\mathrm{d}\xi\right)\mathrm{d}\tau\right\}. \qquad \square
\end{aligned} \tag{3.2}$$

Remark 3.1. The conditions of weak ergodicity of nonhomogeneous continuous-time Markov chains were obtained in [5,10]. Particularly, if the conditions of Theorem 1 in [10] are fulfilled then the respective chain is $(N, \alpha(t))$-ergodic.

Corollary 3.1. *If under the assumptions of Theorem* 3.1 *the Markov chain* $\bar{X}(t)$ *is strongly ergodic with limit regime* Q *(for instance there exists a constant vector* $\boldsymbol{q} \in \Omega$ *such that* $\boldsymbol{q}A(t) = 0$ *for almost all* $t \geqslant 0$*) then* $X(t)$ *is also strongly ergodic with the same limit regime.* □

Similar statements for the case of homogeneous $\bar{X}(t)$ were considered in [6].

Corollary 3.2. *Let* $\bar{X}(t)$ *be a strongly ergodic homogeneous chain* $(\bar{A}(t) \equiv \bar{A})$ *with limit regime* Q*, and let the intensity matrix* $A(t)$ *of the chain* $X(t)$ *be asymptotically constant* $(a(t) \leqslant \alpha\omega(t))$*. Then* $X(t)$ *is strongly ergodic and*

$$\|P(s, t) - Q\| \leqslant 2Ne^{-\alpha(t-s)} \left\{1 + 6\alpha \int_s^t \omega(\tau) e^{\alpha(\tau - s)}\, d\tau\right\}. \qquad \square \tag{3.3}$$

One can obtain a more explicit estimate of the rate of convergence using the following statement.

If $\alpha(t)$ is a non-negative locally integrable function on $[0, \infty)$ and condition (1.9) is satisfied, then there exists a non-decreasing function $\varphi(t)$ such that $\varphi(t) \to \infty$ as $t \to \infty$ and

$$\int_{\varphi(t)}^{t} \alpha(u)\, du \to \infty \quad \text{as } t \to \infty. \tag{3.4}$$

For example one can suppose $\varphi(t) = n$ for $t \in (t_n, t_{n+1})$ where $\int_{t_n}^{t_{n+1}} \alpha(u)\, du = n$.

Let $\omega(t)$ be a non-negative locally integrable function and $\omega(t) \to 0$ as $t \to \infty$. Then the function

$$\psi(t) = \operatorname*{ess\,sup}_{\tau \in [n, n+1)} |\omega(\tau)| \quad \text{if } t \in [n, n+1),\ n \in \mathbb{N}, \tag{3.5}$$

decreases to zero as $t \to \infty$ and $|\omega(t)| \leqslant \psi(t)$ for any $t \geqslant 0$.

Theorem 3.2. *Let* $\alpha(t)$, $\omega(t)$ *be non-negative and such that condition (1.9) is fulfilled, with* $\omega(t) \to 0$ *as* $t \to \infty$*. Then the following estimate is true: for any* $s \geqslant 0$, $\varphi(t) \geqslant s$,

$$\int_s^t \omega(\tau)\alpha(\tau) \exp\left(-\int_\tau^t \alpha(\xi)\, d\xi\right) d\tau$$

$$\leqslant \psi(\varphi(t)) + \psi(s) \exp\left\{-\int_{\varphi(t)}^{t} \alpha(\tau)\, d\tau\right\} \tag{3.6}$$

and therefore the left-hand side of inequality (3.6) tends to zero as $t \to \infty$.

Proof. One has for any $s \geqslant 0$, and $\varphi(t) \geqslant s$,

$$\int_s^t \omega(\tau)\alpha(\tau)\exp\left(-\int_\tau^t \alpha(\xi)\,d\xi\right)d\tau$$

$$\leqslant \exp\left(-\int_0^t \alpha(\xi)\,d\xi\right)\int_s^t \psi(\tau)\alpha(\tau)\exp\left(\int_0^\tau \alpha(\xi)\,d\xi\right)d\tau$$

$$\leqslant \exp\left(-\int_0^t \alpha(\xi)\,d\xi\right)\left\{\psi(s)\int_s^{\varphi(t)} \alpha(\tau)\exp\left(\int_0^\tau \alpha(\xi)\,d\xi\right)d\tau\right.$$

$$\left.+\psi(\varphi(t))\int_{\varphi(t)}^t \alpha(\tau)\exp\left(\int_0^\tau \alpha(\xi)\,d\xi\right)d\tau\right\}$$

$$\leqslant \exp\left(-\int_0^t \alpha(\xi)\,d\xi\right)\left\{\psi(s)\exp\left(\int_0^{\varphi(t)} \alpha(\tau)\,d\tau\right)+\psi(\varphi(t))\int_0^t \alpha(\tau)\,d\tau\right\}$$

$$=\psi(\varphi(t))+\psi(s)\exp\left\{-\int_{\varphi(t)}^t \alpha(\tau)\,d\tau\right\}. \qquad \square \tag{3.7}$$

Now we consider the condition

$$\int_0^\infty a(t)\,dt<\infty \tag{3.8}$$

instead of (2.1). In this case the condition of $(N, \alpha(t))$-ergodicity is unnecessary.

Theorem 3.3. *Let condition (3.8) be fulfilled and let the Markov chain $\bar{X}(t)$ be weakly ergodic. Then the Markov chain $X(t)$ is also weakly ergodic and their limit regimes are identical.*

Proof. From the forward Kolmogorov systems (2.3) for $X(t)$ and $\bar{X}(t)$ one has

$$d\bar{\boldsymbol{x}}/dt = B(t)\bar{\boldsymbol{x}} + (\bar{B}(t) - B(t))\bar{\boldsymbol{x}}\,, \tag{3.9}$$

and if $\bar{\boldsymbol{x}}(s) = \boldsymbol{x}(s)$ then

$$\bar{\boldsymbol{x}}(t) = U(t, s)\boldsymbol{x}(s) + \int_s^t U(t, \tau)(\bar{B}(\tau) - B(\tau))\bar{\boldsymbol{x}}(\tau)\,d\tau$$

$$= \boldsymbol{x}(t) + \int_s^t U(t, \tau)(\bar{B}(\tau) - B(\tau))\bar{\boldsymbol{x}}(\tau)\,d\tau\,. \tag{3.10}$$

Next

$$\|\bar{x}(t)-x(t)\| \leqslant \int_s^t \|U(t,\tau)\|\|\bar{B}(\tau)-B(\tau)\|\|\bar{x}(\tau)\|\,\mathrm{d}\tau$$

$$\leqslant \int_s^t \|\bar{B}(\tau)-B(\tau)\|\,\mathrm{d}\tau = \int_s^t a(\tau)\|\,\mathrm{d}\tau, \tag{3.11}$$

and hence

$$\|P(s,t)-\bar{P}(s,t)\| \leqslant \int_s^t a(\tau)\,\mathrm{d}\tau. \tag{3.12}$$

Now let $\bar{X}(t)$ be weakly ergodic with a limit regime $Q(t)$. Let $\varepsilon>0$ be fixed. There exists s such that $\int_s^\infty a(\tau)\,\mathrm{d}\tau<\frac{1}{2}\varepsilon$. One can choose t_0 such that $\|\bar{P}(s,t)-Q(t)\|<\frac{1}{2}\varepsilon$ for $t\geqslant t_0$. Then using (3.12) one has

$$\|P(s,t)-Q(t)\| \leqslant \|P(s,t)-\bar{P}(s,t)\|+\|\bar{P}(s,t)-Q(t)\|$$

$$\leqslant \int_s^t a(\tau)\,\mathrm{d}\tau+\tfrac{1}{2}\varepsilon<\varepsilon. \tag{3.13}$$

Then

$$\|P(0,t)-Q(t)\| \leqslant \|P(0,s)\|\|P(s,t)-Q(t)\|$$

$$\leqslant \|P(s,t)-Q(t)\|<\varepsilon. \tag{3.14}$$

Inequality (3.14) implies weak ergodicity of $X(t)$ and the equality of the limit regimes because $\varepsilon>0$ is arbitrary. □

Corollary 3.3. *If under the assumptions of Theorem 3.3 the Markov chain $\bar{X}(t)$ is strongly ergodic with limit regime Q then $X(t)$ is also strongly ergodic with the same limit regime.* □

4. Queue-length process for M(*t*)/M(*t*)/*N*/0 queue

Let $X(t)$ be a queue-length process for a $\mathrm{M}(t)/\mathrm{M}(t)/N/0$ queue (there are N servers and no waiting rooms, see, for example, [8]). It is a birth and death process with the state space $S=\{0,1,\ldots,N\}$ and intensities $\lambda_{n-1}(t)=\lambda(t)$, $\mu_n(t)=n\mu(t)$, $n=1,\ldots,N$.

There have been a number of investigations of this process in some particular cases, see [2,4,7].

Let the intensities be asymptotically proportional:

$$\lambda(t)/\mu(t)\to\phi \quad \text{as } t\to\infty. \tag{4.1}$$

Then one can write

$$\lambda(t) = \phi\mu(t)\{1+\omega(t)\} \quad \text{where } \omega(t) \to 0 \text{ as } t \to \infty . \tag{4.2}$$

Here $\omega(t)$ can be supposed to be positive for any $t \geqslant 0$ and to be decreasing. Also assume that

$$\int_0^\infty \mu(t)\,\mathrm{d}t = \infty . \tag{4.3}$$

It is noteworthy that if condition (4.3) is not fulfilled then the process is not weakly ergodic.

Let $\bar{X}(t)$ be a birth and death process with proportional intensities

$$\bar{\lambda}_{n-1}(t) = \phi\mu(t) , \quad \bar{\mu}_n(t) = n\mu(t) , \qquad n = 1,\ldots, N . \tag{4.4}$$

Then condition (4.3) implies strong ergodicity of $\bar{X}(t)$ by Theorem 2 of [10], see also [9]; let $\boldsymbol{q}$ be the steady-state distribution of $\bar{X}(t)$. The process $X(t)$ with intensities $\lambda(t)$ and $\mu(t)$ is a perturbation of $\bar{X}(t)$ in the sense of this paper. Hence we can prove ergodicity of $X(t)$ with limit regime $\boldsymbol{q}$ and can estimate the rate of convergence. Let $\boldsymbol{p}(t)$ be the probability distribution for $X(t)$.

Theorem 4.1. *Let the intensities of $X(t)$ be asymptotically proportional and conditions (4.2), (4.3) be satisfied. Then $X(t)$ is strongly ergodic with the limit regime $\boldsymbol{q}$. Moreover for any $s \geqslant 0$, $t \geqslant 0$ and any $\boldsymbol{p}(s) = \boldsymbol{p}$* the following estimate holds:

$$\|\boldsymbol{p}(t) - \boldsymbol{q}\| \leqslant 8N \exp\left(-\int_s^t \mu(\tau)\,\mathrm{d}\tau\right)$$

$$\times \left(1 + 6\phi \int_s^t \omega(\tau)\mu(\tau) \exp\left(\int_s^\tau \mu(\xi)\,\mathrm{d}\xi\right)\mathrm{d}\tau\right). \tag{4.5}$$

Proof. Let $X'(t)$ be the homogeneous birth and death process with state space S, probability distribution $\boldsymbol{p}'(t)$ and intensities

$$\lambda'_{n-1} = \phi , \quad \mu'_n = n , \qquad n = 1,\ldots, N . \tag{4.6}$$

Such a process was considered in [11], Example 2, estimate (14). One has

$$\|\boldsymbol{p}'(t) - \boldsymbol{q}\| \leqslant 8N \exp(-t) . \tag{4.7}$$

Then by Theorem 2 of [10],

$$\|\bar{\boldsymbol{p}}(t) - \boldsymbol{q}\| \leqslant 8N \exp\left(-\int_s^t \mu(\tau)\,\mathrm{d}\tau\right), \tag{4.8}$$

and $\bar{X}(t)$ is $(\mathcal{N}, \alpha(t))$-ergodic with $\mathcal{N}=4N$, $\alpha(t)=\mu(t)$. Instead of (2.1) one has $a(t)=\phi\mu(t)\omega(t)$ and our claim follows from Theorem 3.1. □

Remark 4.1. Ergodicity of this process was proved by D. Gnedenko [4] without any estimates. The results of [3] show that the condition of asymptotic proportionality of intensities is close to a necessary condition for ergodicity.

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